

Hydrogenic C++ Documentation

This note gives a compact overview of the C++ part of the Hydrogenic-demo workspace. The code builds a small executable, `hydrogenic_atom_demo`, that exercises the hydrogenic atom routines in `Hydrogenic.vX` together with numerical utilities from `Tools`.

Project layout

Path	Purpose
<code>main.cpp</code>	Example program that constructs hydrogenic atoms and prints representative atomic data.
<code>Hydrogenic.vX/Atom.*</code>	Core <code>Atom</code> , <code>Atomic_Shell</code> , <code>Electron_Level</code> , and <code>Gas_of_Atoms</code> classes.
<code>Hydrogenic.vX/Oscillator_strength.*</code>	Bound-bound oscillator strength and transition support.
<code>Hydrogenic.vX/Photoionization_cross_section.*</code>	Photoionization cross sections and Gaunt factors.
<code>Hydrogenic.vX/Quadrupole_lines_HI.*</code>	Electric quadrupole line rates for hydrogenic atoms.
<code>Hydrogenic.vX/Rec_Phot_BB.*</code> and <code>Recombination_Integrals.*</code>	Recombination and photon interaction helpers.
<code>Tools/</code>	Shared constants, integration routines, line profiles, and simple numerical helpers.
<code>Makefile</code> and <code>Makefile.in</code>	Build entry points for the demo executable.

What the C++ example does

The demo constructs hydrogenic atoms with a selected number of complete shells. Each shell contains angular momentum levels with $l = 0 \dots n - 1$. The code can report:

- level degeneracies, ionization energies, ionization frequencies, total decay rates, and lifetimes;
- selected bound-bound transition frequencies, wavelengths, and Einstein A21 coefficients;
- electric dipole transitions and optional electric quadrupole transitions;
- photoionization cross sections and Gaunt factors at frequencies above threshold;
- simple rescaling behavior for the fine-structure constant and electron mass.

The current `main.cpp` includes two examples:

- hydrogen with $Z = 1$, 5 shells, quadrupole lines enabled, and Storey-Hummer recombination interpolation enabled;
- hydrogenic oxygen, O VIII, with $Z = 8$, 30 shells, quadrupole lines enabled, and Storey-Hummer recombination interpolation enabled.

Scientific background

The hydrogenic atom setup was developed for detailed calculations of cosmological recombination. This requires multi-shell hydrogenic atoms with resolved angular momentum sub-states, dipole transitions, quadrupole transitions, photoionization cross sections, and recombination coefficients. The C++ example in `main.cpp` cites the main background papers for this use case.

References

- [1] J. Chluba and R. M. Thomas, "Towards a complete treatment of the cosmological recombination problem", MNRAS 412, 748, 2011. <https://ui.adsabs.harvard.edu/abs/2011MNRAS.412..748C/abstract>
- [2] J. Chluba, J. A. Rubino-Martin, and R. A. Sunyaev, "Cosmological hydrogen recombination: populations of the high level sub-states", MNRAS 374, 1310, 2007. <https://ui.adsabs.harvard.edu/abs/2007MNRAS.374.1310C/abstract>
- [3] J. Chluba and Y. Ali-Haïmoud, "CosmoSpec: Fast and detailed computation of the cosmological recombination radiation from hydrogen and helium", MNRAS 456, 3494, 2016. <https://ui.adsabs.harvard.edu/abs/2016MNRAS.456.3494C/abstract>

Dependencies

The build expects:

- a C++17 compiler with OpenMP support;
- GSL headers and libraries;
- Boost headers;
- GNU make.

On macOS with Homebrew, the typical setup is:

```
brew install gcc gsl boost
```

`Makefile.in` currently defaults to:

```
CXX = /opt/homebrew/bin/g++-16
DEF_INC_PATH = /opt/homebrew/include
DEF_LIB_PATH = /opt/homebrew/lib
```

If your compiler or Homebrew prefix differs, override these variables when invoking `make`.

Build and run

From the workspace root:

```
make
```

Run the demo:

```
make run
```

Clean generated build products:

```
make clean
```

Update submodules if the project is checked out with submodule dependencies:

```
make update-modules
```

Example build override:

```
make CXX=/opt/homebrew/bin/g++-16 DEF_INC_PATH=/opt/homebrew/include  
DEF_LIB_PATH=/opt/homebrew/lib
```

The executable is written to:

```
./hydrogenic_atom_demo
```

Intermediate object and dependency files are written under:

```
./build/
```

Python and notebooks

This workspace also has a planned Python part. The Python material will live in a python/ area and use Jupyter notebooks to further illustrate the same physics: energy levels, line series, photoionization behavior, recombination examples, and comparisons with the C++ output. Those notebooks are intended as exploratory and visualization-oriented companions to the compiled C++ routines.

Regenerating this PDF

The PDF version of this document can be regenerated from the Markdown source with Pandoc and Typst:

```
pandoc Documentation/Hydrogenic_cpp_Documentation.md --from gfm --pdf-  
engine=typst --output Documentation/Hydrogenic_cpp_Documentation.pdf
```